AeroDNA

01 :

Abstract and Research Question

Decoding the Performance DNA of Formula 1 Cars Through Telemetry

Project report | 2024 qualifying data | Monaco, Monza and Silverstone

AeroDNA investigates how Formula 1 car performance can be explored through telemetry-derived features across circuits with different characteristics. The project uses qualifying data from the 2024 Monaco, Italian and British Grands Prix to examine relationships between lap time, maximum speed, distance-weighted mean speed, high-speed distance percentage and estimated acceleration and deceleration.
Using Python and FastF1, telemetry was extracted from the fastest accurate lap selected for each driver at each circuit. The resulting dataset contains 59 driver–circuit observations across three circuits. Exploratory analysis included correlation matrices, circuit-specific comparisons, clustering, regression baselines and team-level feature comparisons.

A key finding was that the relationship between lap time and distance-weighted mean speed changed direction when observations from different circuits were pooled. The pooled correlation was positive (+0.61), while circuit-specific correlations were strongly negative, ranging from −0.92 to −0.99. This illustrates why circuit context must be considered when interpreting performance data.
The analysis also revealed substantial redundancy between mean speed and high-speed distance percentage, alongside potentially influential laps at Silverstone. These findings demonstrate both the value and the limitations of telemetry-based comparisons. AeroDNA does not directly measure aerodynamic forces or efficiency; instead, it establishes an exploratory framework for investigating performance patterns and identifying questions for more detailed engineering analysis.

02:

Background and Context

Decoding the Performance DNA of Formula 1 Cars Through Telemetry

Project report | 2024 qualifying data | Monaco, Monza and Silverstone

Formula 1 performance emerges from the interaction of vehicle design, aerodynamic configuration, power delivery, tyre behavior, braking, driver input and circuit characteristics. A car optimized for one circuit may not demonstrate the same relative strengths at another. For example, a circuit with long straights and heavy braking zones presents a different performance challenge from one dominated by slow corners and limited opportunities to reach maximum speed.

Telemetry provides a way to investigate these differences through measurable quantities such as speed and acceleration. However, individual telemetry features do not necessarily reveal the underlying engineering explanation. Maximum speed, for instance, reflects multiple influences, including aerodynamic drag, power delivery, gearing, and the conditions in which the measurement was recorded.

AeroDNA approaches this problem as an exploratory data-analysis project. Rather than claiming to reconstruct the aerodynamic characteristics of individual cars from publicly available telemetry, it investigates whether telemetry-derived features reveal distinguishable performance patterns across circuits and teams.

Research question

How can telemetry-derived features be used to investigate circuit-dependent performance patterns in Formula 1 qualifying, and what are the limitations of interpreting these patterns as indicators of vehicle aerodynamics?

The project focuses on three 2024 qualifying sessions: Monaco, Monza and Silverstone. These circuits provide contrasting contexts for examining how observed performance changes across different layouts.

03: Executive Findings

Can telemetry reveal recurring performance patterns ?

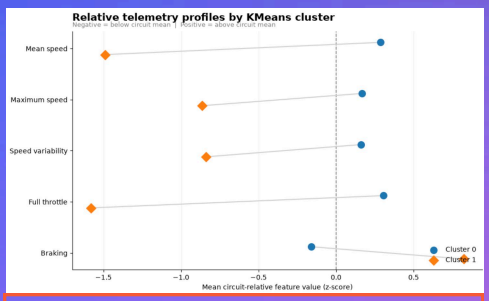
Observations 309

Reference clusters 2

Assigned to cluster 0 83.8%

3 Circuit contexts 3

Findings 01: What separates the two k-MEANS clusters?



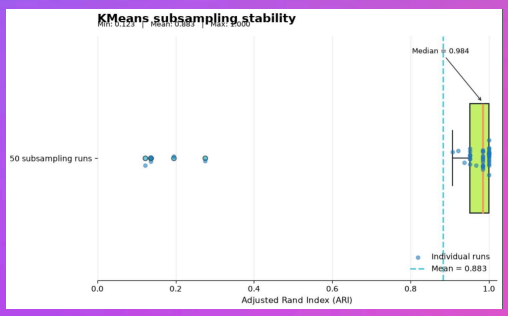
Feature	Cluster 0	Cluster 1
Mean speed	+0.29	−1.49
Maximum speed	+0.17	−0.87
Speed variability	+0.16	−0.84
Full throttle	+0.31	−1.58
Braking	−0.16	+0.82

The k-means reference partition separated observations into two profiles with contrasting relative telemetry characteristics. Cluster 0 tended to sit above it's circuit mean for speed, speed variability and full throttle percentage, while showing slightly lower than average braking values

Cluster 1 displayed the opposite pattern, with lower relative speed and full throttle values and higher relative braking.

This contrast suggests that the model captured a broad difference in the recorded telemetry. However the profiles should not be interpreted as relative groupings rather than fixed categories of driver ability or vehicle performance.

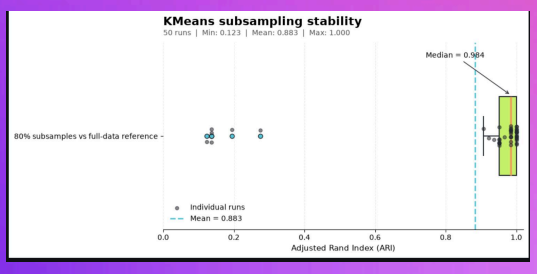
Findings 02: How much confidence can we place in the apparent structure?



Cluster 1 accounted for 19.0% of Monaco observations, 6.9% of Monza observations and 21.2% of Silverstone observations. The profile was therefore much less prevalent in Monza sample than the other two circuits. This uneven distribution suggests that the relative telemetry groupings were not represented uniformly across the data set.

These differences on their own do not establish why cluster proportions varied . Circuit characteristics , the composition of the sampled laps and other factors may contribute.

Findings 03: K - means sub sampling stability

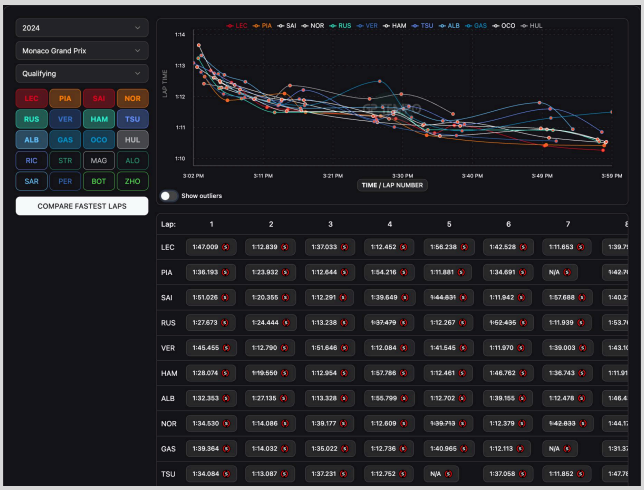


The sub sampling experiement showed that K-means partition often similar to the full data reference, with a median adjusted rand index of 0.984 across 50 runs. However the minimum ARI of 0.123 indicates that some sub sampled produced substantially differ partitions. The results therefore suggests that the identified structure was often reproducible under this procedure, but not consistently so.

Cross circuit transfer was also uneven in the leave one circuit out experiements . This was because each held out circuit was standardised using it's own observations, these experiements assess transfer in circuit relative feature space rather than performance on a completely unseen circuit without prior callibration. Taken together the results support further investigation of the telemetry patterns .

04: Research Context

Why look for Performance Signatures in F1 Telemetry





Digram 01: K-means telemetry comparison : The dashboard compares two clusters with contrasting telemetry profiles across speed, throttle and braking characteristics. These differences illustrate how k means can group observations according to similarities in their measured features, providing an exploratory view of performance patterns. However the profiles should not be interpreted as definitive indicators of driver ability or vehicle superiority.

Diagram 02: Circuit layout and telemetry traces : The circuit map and corresponding telemetry traces illustrate how performance measures vary through laps. Changes in speed and related telemetry signals reflect the demands of different track sections, including corners and straights. This highlights the importance of circuit context when interpreting performance signatures, as observed patterns may be driven by track layout rather than team specific characteristics.

05: Defining the investigation

Four Research Objectives

01.

02.

03.

04.

Success criterion:
interpretability + stability + cautious cross circuit evaluation
The aim is not to produce the largest possible number of clusters, but to determine whether the resulting structure is interpratable and sufficiently robust to warrant further investigation

Data Set :

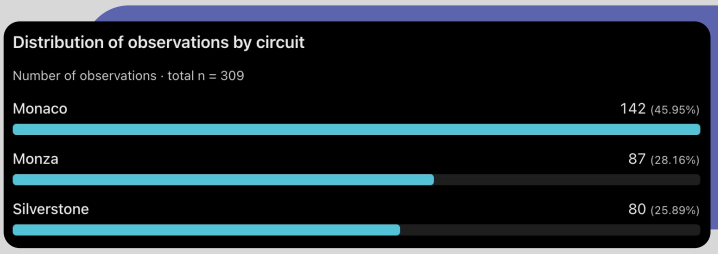
309

2024

3

5

Circuit Selection and Observation Distribution



Monaco: low speed technical circuit. It's tight corners and limited opportunity for sustained high speed running provide a contrasting context for examining speed and braking characteristics.

Monza: High speed straights dominated circuit with heavy braking zones that create a different balance for speed, throttle application and deceleration.

Silverstone: High speed varied circuit -combination of fast corners, straights, and directional changes introduces a different mix of sustained speed.

What does one observation represent

one observation = one recorded driver lap

Each observation is represented by five numerical features

1. Mean Speed km/hr
2. Maximum speed km/hr
3. Speed variability km.hr
4. Full throttle percentage
5. braking percentage

What the data set does not capture

The data set captures select qualifying observations from three circuits in a single season. It does not represent every lap, circuit, session or season in formula 1 and the number of observations differs between circuits.

The numerical features provide a reduced description of recorded vehicle behavior. They do not by themselves account for all factors that can influence telemetry including tyre condition, fuel load, weather, setup choices, traffic or driver strategy. Unless these variables are explicitly included and controlled for, the analysis cannot attribute a cluster to any single cause.

Accordingly the resulting clusters are interpreted as groupings of the available telemetry features, rather than direct measurements of driver skill, car quality or overall competitive performance.

06

Feature Selection and Engineering

Translating telemetry into measurable behaviour

To investigate recurring patterns in formula 1 telemetry, the analysis represents each observation through five numerical features describing different aspects of vehicle behavior. Together these variables provide a simplified representation of how speed and driver inputs vary within recorded data. The features were selected to capture complementary aspects of telemetry while maintaining consistent numerical features set across Monaco, Monza and Silverstone

Feature	What it represents	Why it matters to the investigation
MeanSpeed_kmh	Average speed, in km/h, across the observation.	Represents the overall speed profile within the recorded data.
MaxSpeed_kmh	Maximum recorded speed, in km/h.	Captures the upper end of the observed speed range.
SpeedVariability_kmh	Variation in recorded speed, expressed in km/h.	Helps distinguish observations with steadier speed profiles from those with greater speed variation.
FullThrottle_pct	Percentage of the observation classified as full throttle.	Represents how much of the recorded interval was spent at full throttle.
Braking_pct	Percentage of the observation classified as braking.	Represents the proportion of the interval associated with braking, according to the dataset's calculation.

Monza. km/hr

Why use these features together ?

No single feature provides a complete description of vehicle behavior. Mean and maximum speed capture different aspects of the speed profile, while speed variability describes how much speed changes within the observation. Full throttle and braking percentages add information about recorded control inputs.

By combining these variables, the clustering analysis can examine whether observations with simillar combinations of speed and control input characteristics form recurring groups. These groupings can be identified as telemetry profiles, rather than definitive categories of driving skill, vehicle quality or overall performance.

07

Pre-processing and circuit relative standardisation

The five selected telemetry features differ in units, numerical ranges and distributions across circuits. For example, mean speed at Monza is substantially higher than at Monaco, reflecting differences in circuits and their driving demands. If these raw values were used directly, clustering could be strongly biased by absolute circuit characteristics rather than relative telemetry profiles being investigated.

To address this, each numerical feature was standardized separately within each circuit using z score standardization. This expresses each observation relative to the mean and standard deviation of it's own circuit, allowing the analysis to examine whether observations share similar patterns relative to their respective circuit environments.

Raw Telemetry

164.65
monaco. km/hr

257.66
Monza. km/hr

Standardized Feature Matrix (ready for clustering)

standardised telemtry
Relative feature values
Each value describes how far an observation sits above or below it's circuit feature mean

z-score formula :

$$z = \frac{x - \mu_{\text{circuit}}}{\sigma_{\text{circuit}}}$$

0	+1	-1
at the circuit mean	One SD above	One SD below

Why standardize separately by circuit

-Reduces scale dominance: features measured in km/hr do not automatically outweigh percentages simply because their numericall values are larger.
- Accounts for circuit context : Observations are represented relative to the typical values and variation of their own circuit
- Supports relative profile comparison : the models can investigate combinations of the above and below circuit average features.

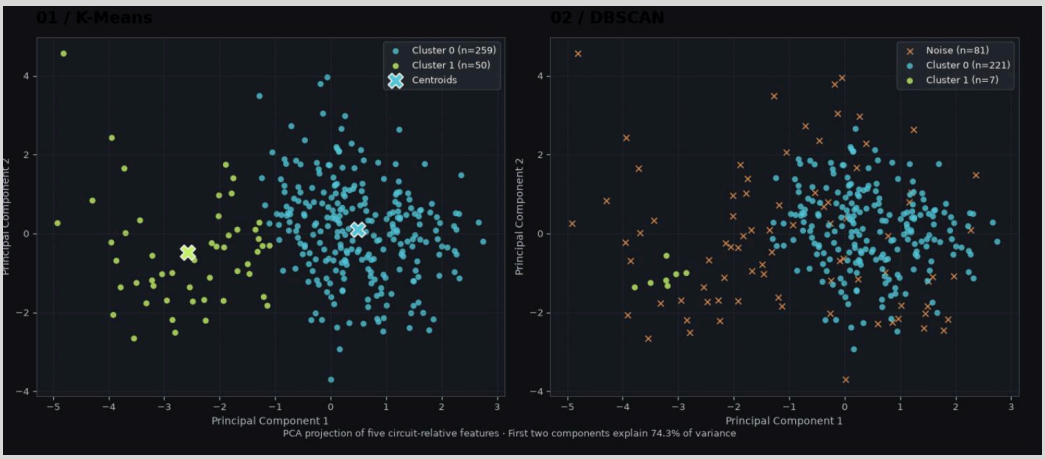
What standardization changes

Circuit relative standardization deliberately removes each circuits original features mean and scale.A standardized speed value of +1 at Monaco and Monza means one standard deviation above the respective circuit mean - not the same absolute mean in km/hr.
As a result this approach investigates relative telemetry patterns, not whether observations across circuits demonstrate equivalent absolute performance. It also means circuit specific differences in variability are normalized which can conceal some meaningful differences between tracks.

With Telemetry transformed into common relative feature space, the next stage applies unsupervised learning to investigate whether recurring groups emerge from the combined feature patterns.

Clustering Methodology

AeroDNA uses unsupervised learning to investigate whether recurring telemetry profiles can be identified without assigning pre defined performance categories to observation. Two clustering algorithms were explored thus far: K-MEANS which partitions observations into a specified number of groups and DBSCAN which identifies dense regions of observations while marking points in low density regions as noise. Using both approaches provides tow different perspectives on the structure of the standardized telemetry : one based on partitioning the feature space, and the other on local density. Their outputs are examined as exploratory groupings rather than assumed to represent objectively distinct types of driving behavior.



01: K-means

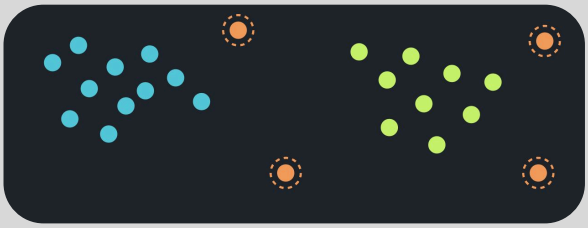


How it works:

1. Select the number of clusters k
2. Assign observations to the nearest centroid
3. Update the centroids and repeat until the solution converges

AERODNA config:
Number of clusters -2
Random State = 42
initialisations: 20

02: DBSCAN



How it works:

1. Define a neighborhood radius epsilon
2. Set the minimum number of nearby observations
3. form density connected clusters and label insufficiently connected observations as noise.
4.

AeroDNA configuration

- Algorithm: DBSCAN
- eps = 1.0
- min_samples = 5
- Input: circuit-relative standardized features

Initial Clustering Outcomes

2 Reference Clusters

259 observations in Cluster 0; 50 in Cluster 1

81 Observations labelled noise

26.2% of 309 observations at eps 1.0 and min_samples 5

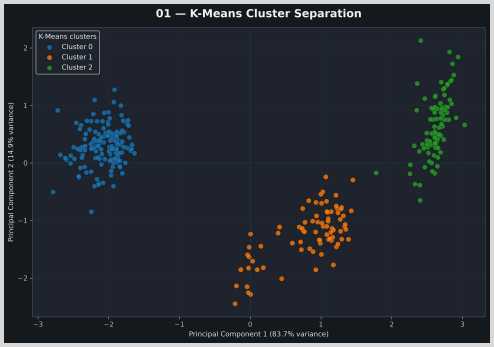
01: K-means

02: DBSCAN

09

Model Results and Cluster interpretation for AeroDNA

01: K-means

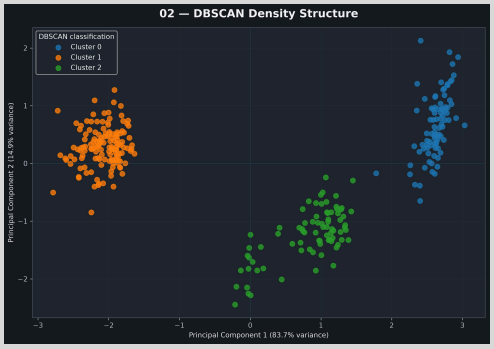


Mapping similarities in telemetry behavior

K-means partitions observations into a pre defined number of clusters according to their similarity across the selected telemetry features . The PCA projection provides a two dimensional view of these groups making it easier to inspect their relative separation and overlap.

The distribution of points indicates how observations relate within the projected feature space. However, visible separation in two dimensions does not, by itself, establish that the clusters represent distinct driving styles or levels of performance.

02: DBSCAN

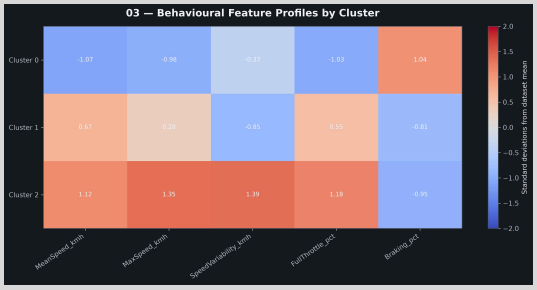


Identifying dense groups and potential outliers

DBSCAN approaches the same features space through density rather than a predefined number of clusters. It groups observations in sufficiently dense regions and labels observations that do not meet it's density criteria as noise.

Comparing this projection with K-means helps investigate whether the data contains irregularly shaped groups or isolated observations that centroid based model may represent differently. The outcome depends on the chosen density parameters so noise labelled observations are not exactly invalid.

03: Feature Profiles

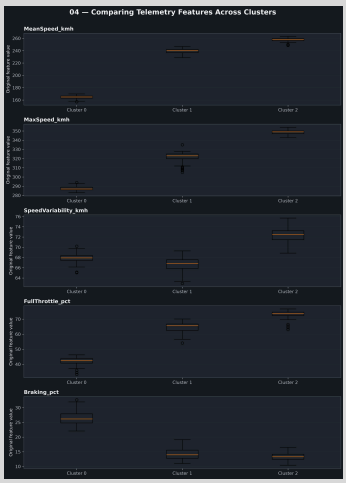


Understanding what differentiates the clusters

The feature heat map compares the mean values of each selected telemetry variable within each k-means cluster. Values are expressed relative to the overall dataset mean in standard deviation units, allowing features measured on different scales to be compared.

Positive values indicate that a clusters mean is above the data sets mean for that feature, while negative values indicate a below average mean.

03: Feature Comparisons

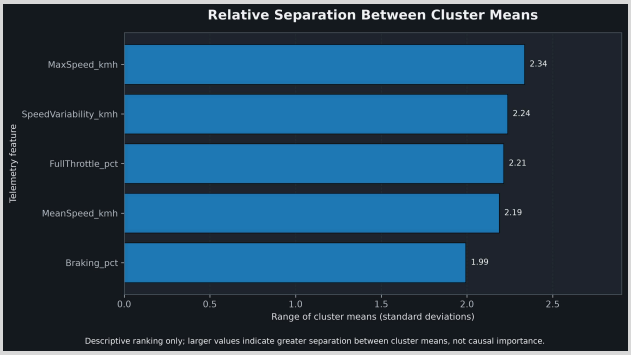


These box plots returns the analysis to original feature units and reveal the distribution of each variable across the k means clusters .Medians, interquartile ranges and outlying observations provide additional context beyond the cluster averages.

Where distributions overlap substantially, differences in the mean profiles may not describe every observation reliably. Reading these distributions alongside the heat map helps distinguish broad tendencies from variation within each cluster.

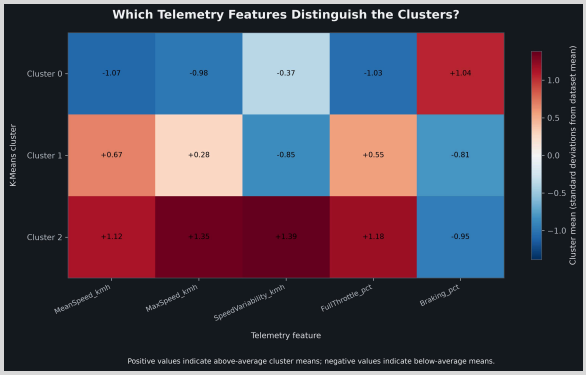
Interpreting the behaviors

01: What separates the clusters



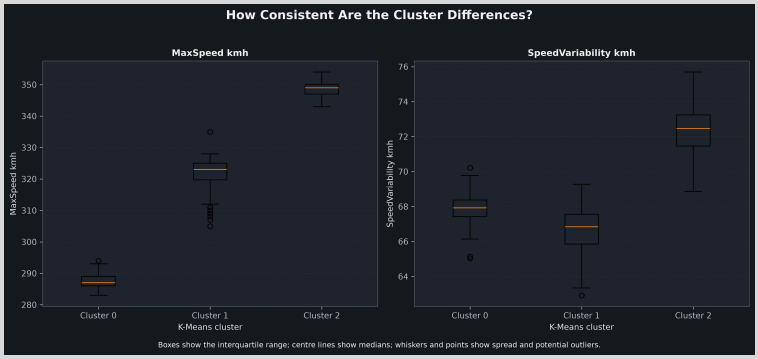
The feature separation ranking identifies which telemetry variables show the largest differences between K means clusters means. This provides a starting point for understanding the numerical characteristics that distinguish the groups, without assuming that any single variable determines overall performance.

02: cluster profiles



The standardized heat map reveals how each cluster's feature means compare with the overall dataset average. Reading these patterns across speed, throttle, braking variables helps identify contrasting telemetry profiles and the features most relevant to further investigation.

03: Distribution and Variation



The distribution plots show whether these differences are consistent across observations or accompanied by substantial variation and overlap. Cluster averages can alone conceal this variation. making the distribution important for judging how broadly each profile describes its observations.

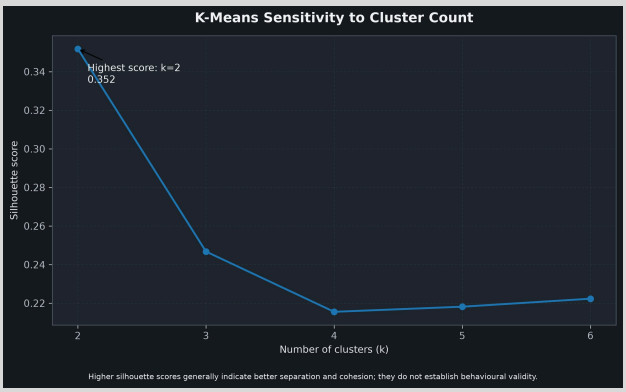
Research insight

Clustering makes similarities in qualifying telemetry easy to investigate, but numerical differences do not independently explain their causes. Circuit characteristics, driver and team context , car setup and session conditions must be considered before drawing conclusions.

Validating the findings

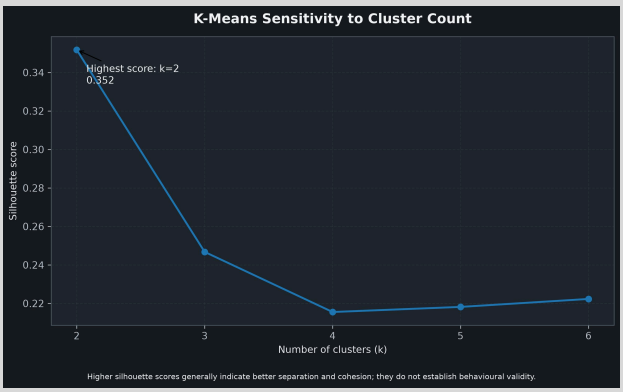
Testing whether the structure identified in qualifying telemetry remains consistent under different model configurations

01: K-means



K means requires the number of clusters to be specific before fitting the model. Comparing silhouette scores across different values of k provides a way to assess how well the resulting groups are separated within the circuit relative feature space. The preferred configuration should be considered alongside cluster sizes and interpretability, rather than selected from a single metric alone.

02: DBSCAN



The dbscan experiments reveal that the identified structure depends substantially on the density threshold . At epsilon = 0.8 , the model grouped the observations into one cluster while labelling 54% as noise. At epsilon = 1 it identified two clusters , with 89/309 observations (26.2%) labelled as noise.Increasing ε to 1.2 and 1.5 again produced one cluster, while reducing the noise proportion to 11.0% and 4.2%, respectively.

03: DBSCAN - the key finding

ε - 0.8	ε = 1.0	ε = 1.5
54%	**26.2%**	**4.2%**
noise	noise	noise

The density threshold substantially changes how DBSCAN classifies the same observations. At ε = 1.0, it identifies two clusters , but one contains only seven observations. This makes cluster balance and parameter sensitivity essential to the interpretation .
These experiments demonstrate that the apparent structure of qualifying telemetry is influenced by the clustering algorithm and it's parameters. DBSCAN did not consistently recover multiple groups across the tested density thresholds, so its output should not be treated as definitive evidence of distinct performance signatures. The K - means sensitivity analysis provides a complementary test of whether a useful grouping emerges across alternative luster counts.

Overall, the validation stage distinguishes patterns that are visible under a particular model configuration from patterns that remain meaningful under alternative settings. Further validation is required before these clusters can be interpreted as repeatable driver behaviors or performance categories.

Discussion, Limitations and implications

The results suggest that unsupervised learning can identify interpretable structure within the selected formula 1 telemetry features , although the meaning and repeatability of that structure require qualification. Using K-means with two clusters , the 309 observations were divided into a larger group of 259 observations (83.8 %) and a smaller group of 50 observations (16.2%). The resulting groups differed in their circuit relative speed, throttle and braking profiles, providing an initial basis for investigating recurring patterns in recorded vehicle behavior.

01: Evidence strength

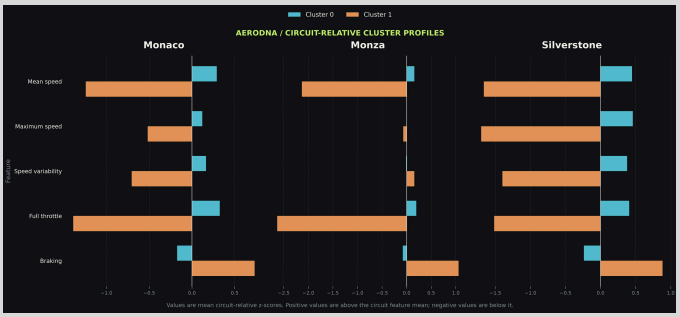
Supported by this analysis

1. two interpretable k-means profiles emerged
2. Profiles differs across relative speeds and input
3. Both clusters contain observations from multiple teams
4. Cluster proportions vary by circuit

Not yet established

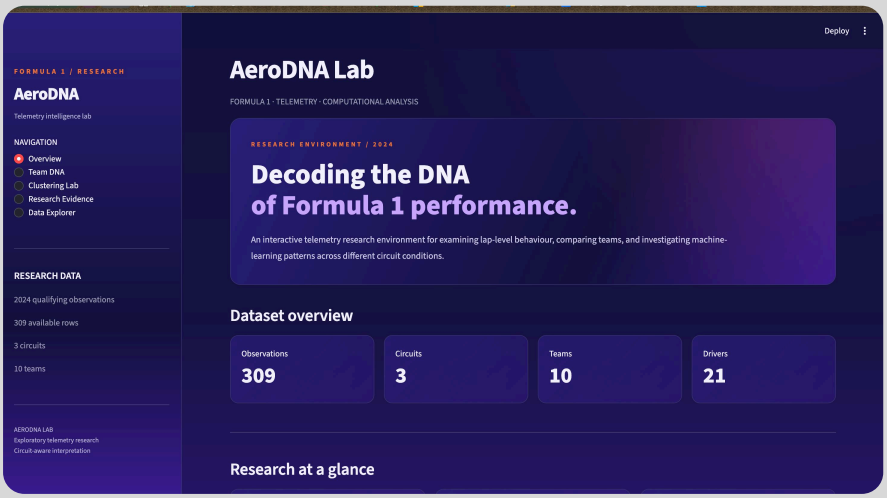
1. Universal profile signatures
2. Causal explanations for the groupings
3. Independent validation on unseen circuits
4. Direct measures of driver skill or car quality.

02: Circuit Level comparison



The circuit level comparison reveals that the relative telemetry profiles vary across Monza, Monaco and Silverstone rather than transferring uniformly . Cluster 1 accounts for 19.0% of the Monaco observations , 6.9% at Monza, and 21.2% at Silverstone. Although the clusters show broadly varying speed, throttle and braking characteristics, these differences in distribution suggest that the identified patterns may be influenced by circuit specific conditions. The results therefore indicate promising relative groupings but not yet universally repeatable performance signatures.

03: Limitations and Implications



The analytical stages of AeroDNA established a foundation for investigating hoe formula 1 qualifying performances varies across circuits and how these differences can be identified through data driven methods. AeroDNA Lab is developed as a interactive research interface using streamLit python and plotly to bring the projects data sets, comparative visualizations and clustering experiments together within a single environment.
The dashboard is organized into five sections Overview, TeamDNA, Clustering Lab, Research Evidence and Data Explorer.



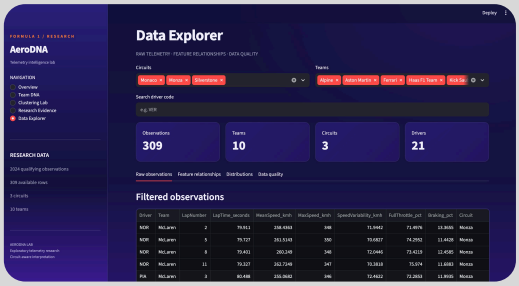
The teamDNA module enables comparative analysis of Formula 1 constructors using qualifying performance indicators.Statistical summaries allow users to examine differences in performance characteristics across teams. This provides a comparative view of the dataset, while recognizing that observed patterns.



The clustering labs presents the saved K-means and DBSCAN clustering results, allowing users to explore how qualifying observations are grouped according to their performance. Users can modify selected features and clustering parameters to investigate alternate groupings.



The data explorer provides direct access to the underlying qualifying data set, enabling users to inspect individual observations, compare performance variables and filter records by relevant categories.



This section consolidates the project's precomputed analytical visualizations, including cluster feature profiles, feature separation rankings, sensitivity analysis and cross circuit comparisons.

Conclusion and Links

This project investigated whether Formula 1 qualifying performance data could be used to identify meaningful patterns across teams and circuits through comparative analysis and unsupervised machine learning. Using 309 qualifying observations from Monaco, Silverstone and Monza during the 2024 season, the analysis explored differences in speed, speed variability, full throttle usage and braking behavior. The clustering results demonstrated a strong separation between the three circuit groups, with the controlled K-means analysis achieving a silhouette score of approximately 0.74. However, because the resulting clusters corresponded directly to circuit identity, the findings primarily demonstrate the influence of circuit dependent performance characteristics rather than independently establishing distinct team engineering philosophies.

Future Scope: Developing AeroDNA into and F1 analytical tool

The next stage of AeroDNA will focus on expanding the data set and refining the analytical framework so that circuit effects can be distinguished more effectively from team and car specific performance patterns . Incorporating additional circuits, qualifying sessions and race data across multiple events and seasons would provide a broader basis for comparison.

Methodologically the project will move beyond relying on cluster separation alone. Future experiments will compare clustering algorithms, investigate the sensitivity of the results to feature selection and scaling, and test whether identified patterns remain stable across circuits, sessions and seasons.

The longer term objective is to develop AeroDNA into an interactive motorsport research platform that supports circuit comparison, performance profiling, pattern discovery and evidence based investigation.

AeroDNA lab : https://aerodna.streamlit.app/

AeroDNA Lab has been developed as a local, interactive dashboard using Python and StreamLit. The source code , analytical notebooks, supporting datasets and research outputs are organized in the project's GitHub repository. Publishing the project provides a transparent record of it's development and enables other users to inspect the methodology, reproduce the analytical workflow where the required data and analytical and workflow where the required data and dependencies are available.

https://github.com/prag3103/AeroDNA